

Sathwik Mohan Vallala

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SUMMARY

Application and Product Security Engineer with 6+ years of experience securing software across the Secure Software Development Lifecycle (Secure SDLC), from threat modeling and secure-by-design reviews through secure code review, CI/CD security, vulnerability management, penetration testing, and incident response. Hands-on experience embedding SAST, DAST, and SCA into DevSecOps workflows; assessing API, Kubernetes, and container security; and building Python security automation. Applies agentic AI and LLM Security techniques to accelerate analysis and remediation, brings practical AWS and cloud security knowledge, and partners with engineering teams to make secure implementation the default. Certified Ethical Hacker (CEH).

EXPERIENCE

Product Security Engineer

Jun. 2024 – Jul. 2026

Omnissa, LLC

Herndon, VA

- Owned Application Security and Product Security risk across 30+ customer features, using OWASP-aligned vulnerability assessments and attack-surface analysis to prioritize threats and recommend practical defenses.
- Embedded secure-by-design practices early in the Secure SDLC through STRIDE threat modeling and security design reviews, identifying attack paths and validating mitigations for 10+ major initiatives.
- Built DevSecOps controls into 20+ releases by incorporating secure code review and high-confidence SAST, DAST, and SCA findings into CI/CD security workflows.
- Led risk-based vulnerability management for 50+ critical findings, applying CVSS to prioritize remediation and validate practical mitigation strategies.
- Applied AI Security, agentic/LLM threat analysis, and AI-assisted penetration testing across 10+ product releases, accelerating reconnaissance, payload generation, and vulnerability triage.
- Assessed API Security and Kubernetes/container risks during design and release reviews, translating findings into actionable engineering mitigations.
- Expanded AI-assisted static analysis and code understanding across 75% of product codebases, identifying risks earlier and scaling secure code review coverage.
- Presented complex security risks to 5+ cross-functional teams and led a monthly cybersecurity roundtable on emerging threats, strengthening secure-development awareness.

Product Security Engineer (Carbon Black)

Nov. 2023 – Jun. 2024

Broadcom Inc.

Herndon, VA

- Identified 10 Product Security vulnerabilities through vulnerability assessment and CVSS-based prioritization, helping engineering teams reduce attack surface and focus remediation on material risk.
- Validated application vulnerabilities through OWASP-aligned penetration testing and DAST across 4 product releases.
- Conducted STRIDE threat modeling and early security architecture reviews to guide pragmatic secure-by-design decisions with engineering teams.
- Performed SAST and SCA informed secure code reviews within CI/CD workflows to identify and mitigate exploitable software and dependency risk.
- Translated complex technical findings into clear risk context, building understanding and remediation buy-in across product teams.

Security Response Engineer

Apr. 2022 – Nov. 2023

VMware Inc.

Herndon, VA

- Owned the vulnerability-management lifecycle for 50+ externally reported findings, performing triage, risk prioritization, remediation coordination, and closure across product teams.
- Conducted vulnerability assessments, assigned CVSS v3.1 severity scores, and documented exploitation paths, mitigations, and compensating controls.
- Partnered with security researchers on responsible disclosure and published 10+ VMware Security Advisories (VMSAs) with clear customer remediation guidance.
- Monitored active exploitation, developed indicators of compromise (IOCs), and updated mitigations and advisories as threat conditions evolved.
- Developed security-response automation that streamlined vulnerability triage, stakeholder coordination, and operational workflows.

Digital Forensic Researcher & Developer

Dec. 2020 – Mar. 2022

ADF Solutions Inc

Reston, VA

- Built 10+ production Python plugins to parse and extract digital artifacts, automating evidence collection and investigative workflows.
- Designed collection approaches for 20+ digital artifacts and independently developed new modules for complex evidence-acquisition use cases.
- Improved module accuracy and reliability through code-level testing, quality assurance, and validation before integration into the operational DEI Tool.

Trainee Security Analyst - Internship

SOC Experts

Jul. 2019 – Oct. 2019

Bangalore, India

- Bolstered security with diverse SIEM tools like Splunk, QRadar, LogRhythm for threat detection response.
- Analyzed monitored logs to uncover suspicious activity potential incidents.
- Conducted basic digital forensics, collecting analyzing evidence for investigations.
- Thrived in simulated SOC, collaborating effectively to identify respond to security threats.

Security Analyst - Internship

Berry9 IT Services Pvt Ltd

Jun. 2017 – Jun. 2018

Hyderabad, India

- Mastered ethical hacking through personal contract projects, fueled by passion.
- Identified vulnerabilities in real-world healthcare pen testing project.
- Earned CEH certification, validating ethical hacking expertise.

EDUCATION

George Mason University

Master of Science in Digital Forensics & Cyber Analysis - 3.89/4.0

Fairfax, VA

Jawaharlal Nehru Technological University

Bachelor of Technology in Computer Science Engineering - 4.0/4.0

Hyderabad, India

PROJECTS

Collaborative ToDo | Node.js, JavaScript, AI-Assisted Development

Apr. 2026 – May 2026

- Built and deployed a collaborative task-management application using AI-assisted software-development workflows and self-hosted infrastructure.
- Implemented secure passwordless authentication and access control to protect shared user data while streamlining access.

Autopsy WhatsApp Plugin | Python, Forensics, Evidence

Apr. 2020 – May 2020

- Enhanced Autopsy capabilities by independently developing a new plugin, demonstrating expertise in forensics tool customization automation.
- Recovered parsed digital evidence from the WhatsApp application on Windows systems, utilizing advanced data extraction parsing techniques.

Chat over network with Python | Python, Network, Encryption

Jun. 2018 – Jul. 2018

- Built a privacy-forward chat application with strong cryptography, empowering secure communication.
- Implemented robust end-to-end encryption for secure communication, ensuring message confidentiality protecting user data
- Enabled user control independence by designing a self-hostable chat server, fostering privacy data sovereignty

File Sharing | Python, Network, Encryption

Jun. 2018 – Jul. 2018

- Streamlined internal file sharing with a user-friendly platform designed for specific network needs.
- Prioritized security by implementing password authentication to restrict unauthorized access to shared files.

CERTIFICATIONS & AWARDS

Security Code Review Ninja - 2nd Degree Black Belt | Secure Coding Dojo

December 2023

Digital Evidence Investigator | ADF Solutions

December 2020

Certified Autopsy User | Basis Technology

October 2020

Certified Ethical Hacker | EC Council

October 2017

TECHNICAL SKILLS

Application & Product Security: Application Security, Product Security, Secure SDLC, Secure-by-Design Architecture, STRIDE Threat Modeling, Secure Code Review, Vulnerability Management, Penetration Testing, API Security, Attack Surface Analysis, OWASP, CVSS v3.1, Incident Response, Responsible Disclosure

DevSecOps & Automation: DevSecOps, CI/CD Security, Static Application Security Testing (SAST), Dynamic Application Security Testing (DAST), Software Composition Analysis (SCA), Security Automation, Python, Bash, Coverity, Semgrep

Cloud, Platform & Detection: AWS Security, Cloud Security, Kubernetes Security, Container Security, Docker, SaaS Security, Linux, Splunk, IBM QRadar, LogRhythm, Log Analysis, Security Monitoring

AI Security: AI Security, Agentic AI Security, LLM Security, AI-Assisted Offensive Security

Offensive Security Tools: Burp Suite, OWASP ZAP, Metasploit, Nmap, Nessus, Wireshark, SQLMap, Nikto

Programming & Additional Security: Python, Bash, Go, JavaScript, SQL, PHP, Digital Forensics, Network Forensics, Memory Forensics, Malware Reverse Engineering, Open Source Intelligence, Ghidra, Volatility, Autopsy, EnCase, FTK